Project Description: Multigraded Homological Algebra and Geometry

Overview. Castelnuovo–Mumford regularity and graded Betti numbers link the geometry of a projective variety to the algebra of its equations. This proposal carries this theory beyond the standard grading: to nonstandard $\mathbb{Z}$-gradings, which arise from stacky curves, and to $\mathbb{Z}^r$-gradings, which arise from toric varieties. After reviewing results of prior NSF support (§1), we develop a Green's-conjecture-type theory for stacky canonical curves (§2); structural results and geometric bounds for multigraded regularity (§3); a uniform regularity theory on the Cox category of a toric variety (§4); and open-source software (§5), with broader impacts in §6.

1. Results of Prior NSF Support

1.1 Intellectual Merit from Prior NSF Support From 2020 to 2022, the PI held an NSF Postdoctoral Research Fellowship (MSPRF DMS-2002239, \$150,000), *Asymptotic Syzygies in Algebraic Geometry*. As publication lags the award, these figures span 2020–2025:: 7 arXiv preprints [38, 40, 39, 28, 27, 7, 41] and 7 published articles, including in *Geometry & Topology* [27], *Journal of Algebra* [38], and *Experimental Mathematics* [42]. This work advanced multigraded homological algebra (§1.2) and the cohomology of moduli spaces (§1.4).

1.2 Eisenbud–Goto on Products of Projective Spaces The Castelnuovo–Mumford regularity of a projective variety $X \subset \mathbb{P}^r$ measures the complexity of X through the vanishing of its cohomology. Its centrality rests on the fact that a single integer controls cohomology vanishing, the degrees of generators and syzygies, and the eventual linearity of resolutions.

Theorem 1.1. [68] *Let $\mathcal{F}$ be a coherent sheaf on $\mathbb{P}^n$ and $M = \bigoplus_{e\in\mathbb{Z}} H^0(\mathbb{P}^n, \mathcal{F}(e))$ the corresponding section module. The following are equivalent:*

(1) M is d-regular in the sense of [119, Lect. 14]*;*
(2) $\beta_{p,q}(M) = 0$ for all $p \geq 0$ and $q > d + p$; and
(3) $M_{\geq d}$ has a d-linear resolution in the sense of [68].

Nearly every use of regularity in the literature, regularity of powers of ideals [59, 104], the Eisenbud–Goto conjecture [116], the construction and boundedness of Hilbert schemes [87, 83], Gröbner basis complexity [13], passes through these equivalences.

A toric variety X is a well-behaved compactification of a torus and, under mild hypotheses, a quotient $X = (\mathbb{C}^n \setminus \mathbb{V}(B))/(\mathbb{C}^\times)^r$, where B is the *irrelevant ideal* of X; for $X = \mathbb{P}^n$, this is $B = \langle x_0, \ldots, x_n\rangle$. Cox's theorem [56] identifies coherent sheaves on X with B-saturated $\mathbb{Z}^r$-graded modules over the *Cox ring* $S = \mathbb{C}[x_1, \ldots, x_n]$, translating the geometry of X into multigraded commutative algebra. This dictionary is far subtler than for $\mathbb{P}^n$, and its subtleties drive much current work: virtual resolutions and resolutions of the diagonal [18, 93, 33, 76], generalizations of the BGG correspondence [67, 35], and multigraded local cohomology [25, 50].

Extending regularity beyond projective space has a long history: Benson's theory for nonstandard $\mathbb{Z}$-graded rings [16], developed by Symonds in his groundbreaking work on invariant theory [130, 131], and Hoffman–Wang's for products of two projective spaces [96]. Maclagan and Smith's cohomological definition [112] subsumes both and is the standard framework on toric varieties. Throughout, X denotes a smooth projective toric variety, so that $\mathrm{Pic}(X) \cong \mathbb{Z}^r$ and the nef cone $\mathrm{Nef}(X)_\mathbb{R} \subset \mathrm{Pic}(X)_\mathbb{R}$ is full-dimensional. We write $\mathrm{Nef}(X) \coloneqq \mathrm{Nef}(X)_\mathbb{R} \cap \mathrm{Pic}(X)$ for the nef monoid and $B_1, \ldots, B_s$ for its unique minimal generators. For an integer vector $\mathbf{v} = (v_1, \ldots, v_t)$ of any length, set $|\mathbf{v}| \coloneqq v_1 + \cdots + v_t$.

Definition 1.2. [112, Def. 6.6] *A coherent sheaf $\mathcal{F}$ on X is* $\mathbf{d}$-regular*, for $\mathbf{d} \in \mathbb{Z}^r$, if, for all $i > 0$,*

$$H^i(X, \mathcal{F}(\mathbf{e})) = 0 \qquad \text{for all} \qquad \mathbf{e} \in L_i(\mathbf{d}) \coloneqq \bigcup_{\mathbf{v}\in\mathbb{N}^s, |\mathbf{v}|=i} \left(\mathbf{d} - \sum_{j=1}^{s} v_j B_j + \mathrm{Nef}(X) \right).$$

The multigraded regularity *of $\mathcal{F}$ is* $\mathrm{reg}(\mathcal{F}) \coloneqq \{\mathbf{d} \in \mathbb{Z}^r \mid \mathcal{F} \text{ is } \mathbf{d}\text{-regular}\}$.

The regularity $\operatorname{reg}(M)$ of a $\mathbb{Z}^r$-graded S-module M is defined by the analogous vanishing of the local cohomology modules $H^i_B(M)$. Multigraded regularity is thus not a single number but an infinite subset of $\mathbb{Z}^r$, whose boundary can be intricate (Figure 1).

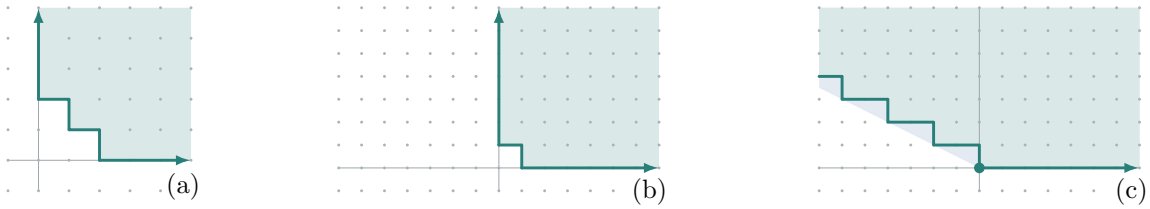


FIGURE 1. Regularity regions of (a) S/I_Z for three points $Z = \{([1:1],[1:4]),([1:2],[1:5]),([1:3],[1:6])\} \subset \mathbb{P}^1 \times \mathbb{P}^1$; (b) $\mathcal{O}_{\mathcal{H}_2}$; (c) S/I_p for a point $p \in \mathcal{H}_2$.

Multigraded regularity has become a central tool in the study of uniform bounds and Hilbert schemes [113, 5], syzygies and local cohomology [127, 94, 25], virtual resolutions [18, 40], and asymptotic regularity [90, 41]. Yet for more than fifteen years after its introduction, a foundational question remained open: *Is there a multigraded Eisenbud–Goto theorem?* That is, can multigraded regularity be characterized by free resolutions and truncations, as in Theorem 1.1? Direct generalizations fail even for products of projective spaces. Answering a question of Maclagan–Smith, further studied by Botbol–Chardin [25], the PI and coauthors showed that multigraded Betti numbers do not determine regularity [40, Ex. 5.1], ruling out an analogue of condition (2). Likewise, the truncation $M_{\geq \mathbf{d}}$ of a $\mathbf{d}$-regular module M need not have a linear resolution, so condition (3) must also be reformulated.

A major advance came with Berkesch, Erman, and Smith's characterization of multigraded regularity on products of projective spaces via virtual resolutions with controlled degrees [18, Thm. 2.9], a counterpart to condition (2). Building on these ideas, the PI and coauthors introduced *quasilinear resolutions* and proved a multigraded Eisenbud–Goto theorem in which quasilinear resolutions of truncations play the role of condition (3).

Theorem 1.3. [40, Thm. A] *Let $\mathcal{F}$ be a coherent sheaf on $X = \mathbb{P}^{n_1} \times \cdots \times \mathbb{P}^{n_r}$ and $M = \bigoplus_{\mathbf{e} \in \mathbb{Z}^r} H^0(X, \mathcal{F}(\mathbf{e}))$ the corresponding section module. For $\mathbf{d} \in \mathbb{Z}^r$, the following are equivalent:*

(1) M is $\mathbf{d}$-regular in the sense of [112]*;*

(2) M has a virtual resolution with degrees as in [18, Thm. 2.9]*; and*

(3) $M_{\geq \mathbf{d}}$ has a $\mathbf{d}$-quasilinear resolution in the sense of [40].

When $r = 1$, Theorem 1.3 recovers the classical Eisenbud–Goto theorem, and (1)⇔(3) holds for every finitely generated M with $H^0_B(M) = 0$. Equally important are its methods: rather than resolving $M_{\geq \mathbf{d}}$ directly, we compute its multigraded Betti numbers via a spectral sequence arising from the Fourier–Mukai transform of $\widetilde{M}$ with Beilinson's resolution of the diagonal [15] as kernel. This "resolution of the diagonal" approach, now a central technique [35, 33, 93, 36], also yields an effective algorithm [3, 58] for testing $\mathbf{d}$-regularity and producing short virtual resolutions.

1.3 Bounds & Asymptotic Behavior of Multigraded Regularity of Ideals Partially order $\operatorname{Pic}(X)$ by $\mathbf{d} \leq \mathbf{e}$ when $\mathbf{e} - \mathbf{d} \in \operatorname{Nef}(X)$. Although nef twists preserve regularity, $\operatorname{reg}(M)$ need not have finitely many minimal elements (pictorially, corners). In [39, Ex. 1.1], the PI, Cranton Heller, and Sayrafi exhibit a cyclic module on a Hirzebruch surface with infinitely many minimal degrees. We show that for faithful modules such pathologies cannot occur.

Theorem 1.4. [39, Cor. 4.6] *Let M be a finitely generated graded S-module with $\operatorname{ann}(M) = 0$ and $\widetilde{M} \neq 0$. If its homogeneous generators f_i satisfy $\mathbf{q} \leq \deg f_i$, then $\operatorname{reg}(M) \subseteq \mathbf{q} + \operatorname{Nef}(X)$. Consequently, $\operatorname{reg}(M)$ has finitely many minimal elements.*

The proof uses the fact that $\mathbf{d}$-regularity forces $M_{\geq \mathbf{d}}$ to be generated in degree $\mathbf{d}$; a chamber-and-determinant argument turns a violation of the bound into a nonzero annihilator of M. It applies to every nonzero homogeneous ideal $I \subseteq S$, though $\operatorname{reg}(S/I)$ may have infinitely many corners.

Classically, the regularity of powers is linearly bounded (Swanson [129]) and eventually linear (Cutkosky–Herzog–Trung, Kodiyalam [59, 104]). Combining our bound with a resolution of the Rees algebra gives the first multigraded analog: regularity of powers translates at a linear rate.

Corollary 1.5. [39, Thm. 5.1] *Let $I = \langle f_1, \dots, f_s \rangle \subseteq S$ be a nonzero homogeneous ideal. There is $\mathbf{a} \in \mathrm{Pic}(X)$ such that for all integers $t > 0$ and all $\mathbf{q}_1, \mathbf{q}_2 \in \mathrm{Pic}(X)$ with $\mathbf{q}_1 \geq \deg f_i \geq \mathbf{q}_2$ for all i,*

$$t\mathbf{q}_1 + \mathbf{a} + \mathrm{reg}(S) \subseteq \mathrm{reg}(I^t) \subseteq t\mathbf{q}_2 + \mathrm{Nef}(X).$$

1.4 Top-Weight Cohomology of $\mathcal{A}_g$ The moduli stack of principally polarized abelian varieties of dimension g, $\mathcal{A}_g \coloneqq [\mathbb{H}_g/\mathrm{Sp}(2g,\mathbb{Z})]$, is one of the most classical objects in algebraic geometry, yet its rational cohomology is known completely only for $g \leq 4$ [99, 91, 97, 98], and until recently little was known beyond Borel's stable range [22]. In particular, Grushevsky asked whether $H^{2k+1}(\mathcal{A}_g;\mathbb{Q}) \neq 0$ for some g and k [88]. Chan, Galatius, and Payne's computation of the top-weight cohomology of $\mathcal{M}_g$, combining Deligne's weight filtration with the combinatorics of tropical moduli spaces, opened a powerful new approach to the cohomology of moduli spaces [47] that is now highly active [48, 122, 20, 44]. The PI and collaborators extended this program to $\mathcal{A}_g$. Since $\mathcal{A}_g$ lacks a modular compactification analogous to $\mathcal{M}_g \subset \overline{\mathcal{M}_g}$, this requires adapting [47] to toroidal compactifications whose boundary strata are indexed by cones in Voronoi's reduction theory. Analyzing these strata and relating the cohomology of $\mathcal{A}_g$ inductively to that of $\mathcal{A}_{g-1}$ and $GL_g(\mathbb{Z})$ yields many new classes.

Theorem 1.6. [27, Thm. A] *The rational cohomology satisfies $H^k(\mathcal{A}_g;\mathbb{Q}) \neq 0$ for $(g,k) = (5,15)$, $(5,20)$, $(6,30)$, $(7,28)$, $(7,33)$, $(7,37)$, and $(7,42)$.*

Theorem 1.6 answers Grushevsky's question affirmatively and, since $H^*(\mathcal{A}_g;\mathbb{Q}) \cong H^*(\mathrm{Sp}_{2g}(\mathbb{Z});\mathbb{Q})$, produces new unstable cohomology for the symplectic group. These results inspired a Hopf algebra structure on the top-weight cohomology of $\mathcal{A}_g$ [30] and Brown's bordification of $\mathcal{A}_g^{\mathrm{Trop}}$ [29], both producing further unstable cohomology of $\mathrm{GL}_n(\mathbb{Z})$. The framework of [27] is the template for the PI's subsequent work on tropicalizing general Shimura varieties [7, 8], the subject of a plenary lecture at the 2025 AMS Summer Research Institute in Algebraic Geometry.

1.5 Broader Impacts from Prior NSF Support

1.5.1 Organizing. During the award, I organized five conferences: *WAGS* (100 participants), *Gems of Combinatorics I* (40), Spec($\overline{\mathbb{Q}}$) (50), *BATMOBILE 2022* (20), and *Gems of Commutative Algebra II* (40), plus AMS special sessions. The *Gems of* series builds early-career research networks.

1.5.2 Mentoring & Virtual Mathematics. I led four undergraduate algebra and geometry reading courses and volunteered with the *Berkeley Math Circle*, promoting mathematics to high schoolers. In Summer 2021, I led six undergraduates in research related to [27] and advised a Summer 2022 project on tropical abelian varieties; that undergraduate is now a mathematics Ph.D. student with an NSF GRF. To reach those at the periphery as mathematics moved online during COVID-19, I helped run Ravi Vakil's massive open-access online course *Algebraic Geometry in the Time of Covid* in Summer and Fall 2020 (∼2,000 participants worldwide) and organized an 8-week virtual undergraduate reading course in commutative algebra in Spring 2021.

1.5.3 Community. During 2020–2023, I served on the board of *Spectra*, including as its inaugural president. I oversaw its growth and formalization, including the adoption of bylaws, the creation of an invited lecture at the JMM, and a fundraising campaign that has raised over $20,000.

2. The Geometry of Syzygies on Stacky Curves

Let X be a projective variety and L a very ample line bundle. For the embedding $X \hookrightarrow \mathbb{P}H^0(X,L) \cong \mathbb{P}^n$ defined by L, let $S_X = S/I_X$, where $S = \mathbb{C}[x_0,\dots,x_n]$ is the standard graded polynomial ring and I_X is the homogeneous defining ideal of X. The minimal graded free resolution $0 \leftarrow S_X \xleftarrow{\epsilon}$

$F_0 \xleftarrow{d_1} F_1 \xleftarrow{d_2} \cdots$ of S_X has terms $F_p = \bigoplus_q S(-q)^{\beta_{p,q}(X;L)}$, where $S(-q)_d = S_{d-q}$. The *graded Betti numbers (or syzygies)* $\beta_{p,q}(X;L)$ depend on the extrinsic embedding, not just on X; a central theme of the field is how they nonetheless capture the intrinsic geometry of X.

Forty years after Green's work, this connection is best understood for smooth non-hyperelliptic canonical curves of genus $g \geq 3$ ($L = K_X$). Green's conjecture [86] predicts the vanishing threshold of the quadratic strand:

$$\beta_{p,p+2}(X; K_X) = 0 \quad \Longleftrightarrow \quad p < \operatorname{Cliff}(X), \qquad 0 \leq p \leq g-3. \tag{1}$$

Here $\operatorname{Cliff}(X) \coloneqq \min\{\deg(L) - 2h^0(X,L) + 2 \,:\, h^0(X,L), h^1(X,L) \geq 2\}$, with the convention $\operatorname{Cliff}(X) = 1$ for a non-hyperelliptic curve of genus 3. Voisin's breakthrough proved it for general curves of every genus [135, 136]; new proofs [4, 125, 101, 103] and refinements [75, 51, 73, 74, 65, 102] followed.

This project develops the corresponding theory for *stacky curves* [2, 43, 1], where S is nonstandard graded and its geometry is visible only through fractional divisors. The classical theory of Noether, Petri, Mumford, and Green is a theory of standard graded rings, and it fails the moment the divisor is not very ample. Already for a point O on an elliptic curve E, the section ring $\bigoplus_n H^0(E, nO)$ needs generators in degrees $1, 2, 3$ and is cut out by the Weierstrass equation $y^2 = x^3 + axz^4 + bz^6$, a sextic in $\mathbb{P}(1,2,3)$. This is not an edge case but the general picture. By Demazure–Pinkham [123, 64, 137], every two-dimensional normal graded domain is $\bigoplus_n H^0(C, \lfloor nD \rfloor)$ for an ample $\mathbb{Q}$-divisor D on a curve C: a section ring on a stacky curve whose stacky points record the fractional part of D. The flagship examples are modular curves; $X(1) = \mathcal{P}(4,6)$ has section ring $\mathbb{C}[E_4, E_6]$, and rings of modular forms on $X_0(N)$ and on Shimura curves are (log) canonical rings of stacky curves, stacky exactly at the elliptic points. Voight and Zureick-Brown [134] proved a stacky Noether–Petri theorem presenting these rings (see also [121, 107]), but their syzygies are uncharted: stacky analogues of Green's N_p theorem and Green's conjecture are wide open. This project builds a theory in which fractional divisors play the role that degree plays classically, extending Mumford's and Green's theorems to stacky curves and answering [34, Ques. 7.7]. Results without citations in this section are proved by the PI joint with M. Banks and P. Wei.

2.1 Weighted Canonical Rings A *stacky curve* $\mathcal{X}$ is a smooth proper geometrically connected Deligne–Mumford stack of dimension one with a dense open subscheme. (Throughout this section, $\mathbb{K} = \mathbb{C}$.) Concretely, $\mathcal{X}$ is a curve C with finitely many *stacky points* carrying cyclic stabilizers [134, Lem. 5.3.10], and the coarse space map $\pi\colon \mathcal{X} \to C$ forgets these stabilizers. If $x_1, \ldots, x_s$ are the stacky points, $e_i = \#G(x_i)$, $P_i = \pi(x_i)$, and $g = g(C)$, the *signature* of $\mathcal{X}$ is $(g; e_1, \ldots, e_s)$. Following [134], the *canonical divisor* of $\mathcal{X}$ is

$$K_{\mathcal{X}} = \pi^* K_C + \sum_i (e_i - 1)\, x_i = \pi^* D \qquad \text{where} \qquad D = K_C + \sum_i \left(1 - \tfrac{1}{e_i}\right) P_i \tag{2}$$

and its degree is $\deg(K_{\mathcal{X}}) = 2g - 2 + \sum_i (1 - 1/e_i)$, which may be a rational number. Thus the genus of $\mathcal{X}$, defined by $\deg(K_{\mathcal{X}}) = 2g_{\mathcal{X}} - 2$, is the rational number $g_{\mathcal{X}} = g + \frac{1}{2}\sum_i (1 - 1/e_i) \geq g$. We say $\mathcal{X}$ is *hyperbolic* when $\deg(K_{\mathcal{X}}) > 0$. The *canonical ring* of $\mathcal{X}$ is

$$R(\mathcal{X}, K_{\mathcal{X}}) \coloneqq \bigoplus_{d \in \mathbb{Z}} H^0\left(\mathcal{X}, dK_{\mathcal{X}}\right) \cong \bigoplus_{d \in \mathbb{Z}} H^0\left(C, \pi_*(dK_{\mathcal{X}})\right) \cong \bigoplus_{d \in \mathbb{Z}} H^0\left(C, \lfloor dD \rfloor\right),$$

where the *round-down* of a $\mathbb{Q}$-divisor $D = \sum_i a_i P_i$ on C is the honest divisor $\lfloor D \rfloor \coloneqq \sum_i \lfloor a_i \rfloor P_i$, and the last isomorphism follows from [134, § 5.4]. When $\mathcal{X}$ is understood, we write R for $R(\mathcal{X}, K_{\mathcal{X}})$.

This round-down is the essential difference from the classical theory. Since $\lfloor D \rfloor = K_C$, $R_1 = H^0(\mathcal{X}, K_{\mathcal{X}}) \cong H^0(C, K_C)$ has dimension $g(C)$ rather than $g(\mathcal{X})$. When stacky points are present, these sections vanish along the ramification divisor $K_{\mathcal{X}} - \pi^* K_C$, so they neither embed $\mathcal{X}$ nor generate R. Adding higher-degree generators yields an embedding into a weighted projective stack [26]; algebraically, R is a quotient of a polynomial ring with a nonstandard $\mathbb{Z}$-grading.

Following Brown and Erman [34, § 3.1], define the *minimal weighted canonical series* $\mathcal{W} = \bigoplus_{d\geq 1} \mathcal{W}_d$ by setting $\mathcal{W}_1 = R_1$ and, for $d \geq 2$, choosing $\mathcal{W}_d$ as a complement in R_d to the degree-d part of the subalgebra generated by $\mathcal{W}_1, \ldots, \mathcal{W}_{d-1}$. Then $\mathcal{W}$ embeds $\mathcal{X}$ into $\mathcal{P}(\mathcal{W})$, and $R \cong S/I_{\mathcal{X}}$, where $S = \mathrm{Sym}(\mathcal{W})$. For $g \geq 2$, the stacky Noether–Petri theorem gives sharp degree bounds of $e := \max(1, e_1, \ldots, e_s)$ for generators and $2e$ for relations, subject to exceptional cases [134, Cor. 1.4.2]. Their inductive presentation also describes relative generators and gives criteria for minimality [134, Thm. 8.3.1(a),(d)]. Theorem 2.1 makes the full weight multiset explicit in our setting. Let $\nu_d = \dim \mathcal{W}_d$; the *weight multiset* of $\mathcal{W}$ has ν_d copies of each weight d, written d^{ν_d}. Set $N = \sum_d \nu_d$, so $\mathbb{P}(\mathcal{W}) = \mathbb{P}(1^{\nu_1}, 2^{\nu_2}, \ldots)$ has dimension $N-1$, $\sigma = \sum_d d\,\nu_d$, and $c = N-2$.

Theorem 2.1. *Let $\mathcal{X}$ be a hyperbolic stacky curve with $g \geq 3$. The weight multiset of $\mathcal{W}$ is $(1^g,\ (2, 3, \ldots, e_1),\ \ldots,\ (2, 3, \ldots, e_s))$, with 2^{g-2} inserted if and only if C is hyperelliptic.*

Thus the weights detect whether C is hyperelliptic, and the generation degree is exactly e. *Proof idea:* Riemann–Roch gives sections $y_{i,j} \in R_j$ with exactly one pole, at P_i of order $j-1$. As every generator of degree m has pole order at most $m-1$, these are indecomposable, and products $\omega \prod_s y_{i,j_s}$ with $\omega(P_i) \neq 0$ realize every admissible pole order. Filter R by pole order at the stacky points: $F_\ell R_d := H^0\big(C, dK_C + \sum_i \min\{\ell, b_i(d)\}P_i\big)$ with $b_i(d) = \lfloor d(e_i-1)/e_i \rfloor$. This multiplicative exhaustive filtration has $F_0 R = R(C, K_C)$. Let $T_e = \mathbb{C}[\Gamma_e]$ for the semigroup $\Gamma_e = \{(n, \ell) : 0 \leq \ell \leq \lfloor n\frac{e-1}{e} \rfloor\}$. The Rees construction flatly degenerates R to $\mathrm{gr}_F R$, separating principal parts at the stacky points.

Theorem 2.2. *Let $\mathcal{X}$ be a hyperbolic stacky curve with $g \geq 3$. Then $\mathrm{gr}_F R$ is the fiber product of $R(C, K_C)$ and $\prod_i T_{e_i}$ over $\prod_i \mathbb{C}[z_i]$, where the maps evaluate coarse sections at P_i and kill positive-pole monomials in T_{e_i}. In particular, there is an exact sequence*

$$0 \longrightarrow \mathrm{gr}_F R \longrightarrow R(C, K_C) \oplus \bigoplus_i T_{e_i} \longrightarrow \bigoplus_i \mathbb{C}[z_i] \longrightarrow 0. \tag{3}$$

Since $H^1(C, nK_C) = 0$ for $n \geq 2$, graded pieces are leading coefficients of principal parts, which multiply monomially at one point and vanish across distinct points. As the generators of Theorem 2.1 generate $\mathrm{gr}_F R$, $\beta(R)$ is $\beta(\mathrm{gr}_F R)$, computed from (3), minus consecutive cancellations.

2.2 The Shape of Stacky Green's Conjecture Classically, the Betti table of a canonical curve has four rows and $g-1$ columns. The top and bottom rows each contain a single 1, so the content lies in the two middle rows, which are exchanged by the Gorenstein duality $\beta_{p,q}(C, K_C) = \beta_{g-2-p,\, g+1-q}(C, K_C)$. Thus Green's conjecture does two things at once: (i) it determines the length of the linear strand of every canonical curve, and (ii) together with the Hilbert function, it determines the entire Betti table of a general curve, since for general C the two middle rows do not overlap. Stacky canonical rings remain Gorenstein, but can have far more than four rows.

Theorem 2.3. *Let $\mathcal{X}$ be a hyperbolic stacky curve with $g \geq 3$. Let $R = S/I$ be a minimal graded presentation, where $S = \mathbb{K}[x_1, \ldots, x_N]$ has total weight σ, and let $c = N-2$.*

(1) R is Gorenstein, with $\beta_{i,j}(R) = \beta_{c-i,\sigma+1-j}(R)$.

(2) The weighted regularity, in the sense of [34]*, of R is* 3*.*

(3) Its nonzero Betti entries lie in rows $0 \leq j-i \leq \rho-1$, where $\rho = \sigma - N + 4$, and both endpoint rows are attained. By Theorem 2.1, $\rho = 4 + \sum_i \binom{e_i}{2}$ if C is non-hyperelliptic and $\rho = g + 2 + \sum_i \binom{e_i}{2}$ if C is hyperelliptic.

Parts (1) and (2) follow from a local cohomology calculation showing that R is a two-dimensional Cohen–Macaulay normal domain with $\mathrm{pd}_S R = c$, together with graded local duality, or alternatively from Watanabe's canonical module formula [137]. The pattern behind part (3) is striking: over a non-hyperelliptic coarse curve, whose canonical Betti table has 4 rows and $g-1$ columns, each stacky point of order e adds $e-1$ columns and $\binom{e}{2}$ rows (Figure 2). The two consequences of classical Green's conjecture thus become separate, new problems in the stacky world.

coarse

	0	1	2	3	4
0	1	–	–	–	–
1	–	6	8	3	–
2	–	3	8	6	–
3	–	–	–	–	1

$(6;2)$

	0	1	2	3	4	5
0	1	–	–	–	–	–
1	–	6	8	3	–	–
2	–	8	24	24	8	–
3	–	–	3	8	6	–
4	–	–	–	–	–	1

$(6;3)$

	0	1	2	3	4	5	6
0	1	–	–	–	–	–	–
1	–	6	8	3	–	–	–
2	–	8	24	24	8	–	–
3	–	6	24	36	24	6	–
4	–	–	8	24	24	8	–
5	–	–	–	3	8	6	–
6	–	–	–	–	–	–	1

FIGURE 2. Betti tables of a genus-6 trigonal canonical curve C and of stacky curves $(6;2)$, $(6;3)$ over C.

Research Problem 2.4. *Let $\mathcal{X}$ be hyperbolic, with non-hyperelliptic coarse space C of genus $g \geq 3$.*

(1) (Weak Stacky Green's Conjecture): Is the linear strand of $\beta(\mathcal{X}; K_{\mathcal{X}})$ determined by $\mathrm{Cliff}(C)$*?*

(2) (Strong Stacky Green's Conjecture): For a sufficiently general stacky curve $\mathcal{X}$, is $\beta(\mathcal{X}; K_{\mathcal{X}})$ determined by $\mathrm{Cliff}(C)$ *and its Hilbert function?*

2.3 Toward a Weak Form of Stacky Green's Conjecture Here and in §2.4, $\mathcal{X}$ and C are as in Research Problem 2.4. The weak form uses Brown–Erman's nonstandard N_p-conditions: $Z \subset \mathbb{P}(\mathcal{W})$ is *weighted* N_p if it is normally generated and F_i is generated in degree at most w^{i+1} for all $i \leq p$, where w^j sums the j largest weights [34, Def. 1.2]; standard weights recover Green's classical N_p. Let $\tau_{\mathrm{BE}}(\mathcal{X})$ be the smallest $p \geq 1$ at which weighted N_p fails. For non-stacky curves, their theorem gives weighted N_p for $1 \leq p \leq \dim \mathcal{W} - g - 2$ for any normally generated weighted series $\mathcal{W}$ of a line bundle of degree $\geq 2g+1$ with $\nu_1 > g$ [34, Thm. 1.7]. The stacky canonical series has $\nu_1 = g$, beyond their methods; [34, Ques. 7.7] requests this canonical analogue. We predict that stacky points shift the coarse threshold by $\kappa := N - g = \sum_i (e_i - 1)$, the number of higher-weight generators. Let $\tau(C, K_C)$ be the first failure of classical N_p. By duality, $\tau(C, K_C) = g - 2 - \ell(C)$, where $\ell(C) := \max\big(\{0\} \cup \{r \geq 1 : \beta_{r,r+1}(C; K_C) \neq 0\}\big)$.

Conjecture 2.5 (Weak Stacky Green's Conjecture)**.** *The threshold is $\tau_{\mathrm{BE}}(\mathcal{X}) = \tau(C, K_C) + \kappa$; equivalently, $\tau_{\mathrm{BE}}(\mathcal{X}) = (N-2) - \ell(C)$.*

For general coarse curves, Voisin's theorem [135, 136] gives $\tau(C, K_C) = \mathrm{Cliff}(C)$, so Conjecture 2.5 predicts $\tau_{\mathrm{BE}}(\mathcal{X}) = N - g + \mathrm{Cliff}(C)$, uniformly over signatures and markings. The same shift motivates Conjectures 2.13 and 2.14. The first ingredient is the following comparison of linear strands.

Proposition 2.6. *For every $r \geq 1$, the inclusion $R(C, K_C) \hookrightarrow R$ induces an isomorphism $K_{r,1}(C, K_C) \xrightarrow{\sim} K_{r,1}(\mathcal{X}, K_{\mathcal{X}})$. In particular, $\beta_{r,r+1}(\mathcal{X}; K_{\mathcal{X}}) = \beta_{r,r+1}(C; K_C)$.*

Proposition 2.6 follows by comparing Koszul cycles and boundaries using minimality of the weighted generators. Together with duality (and the final Betti entry when $\ell(C) = 0$), it gives the nonvanishing half of Conjecture 2.5. For vanishing, let u_k be the sum of the k smallest weights. Duality reduces the problem to $\beta_{r,q}(R) = 0$ for $\ell(C) < r < c$ and $q \leq u_{r+1}$. Comparison and minimality settle $r < g$. I will begin with one order-four point, where the only new requirement is $\beta_{g,g+2} = 0$, equivalently $\beta_{1,8} = 0$. The mapping cone of (3), using the semigroup relations of T_4, will test this on the special fiber and isolate any cancellation needed after deformation. For arbitrary signatures, I will insert the weighted Eagon–Northcott resolutions of the regraded rational-normal-curve rings T_{e_i} (see [63, 12]) into the fiber-product mapping cone and bound its surviving shifts. Showing that every surviving shift in homological degree $g \leq r < c$ exceeds u_{r+1} would complete the vanishing argument, proving Conjecture 2.5.

Finally, gonality and Clifford index supply no additional stacky invariant (here $\mathrm{Cliff}(\mathcal{X})$ is defined using line bundles with $h^0, h^1 \geq 2$, with the usual low-genus conventions).

Lemma 2.7 (Brill–Noether rigidity)**.** *We have* $\operatorname{Cliff}(\mathcal{X}) = \operatorname{Cliff}(C)$ *and* $\operatorname{gon}(\mathcal{X}) = \operatorname{gon}(C)$*. Every line bundle attaining either minimum is a pullback.*

Indeed, write $\mathcal{L} = \pi^*M \otimes \bigotimes_i \mathcal{T}_i^{a_i}$, where $\mathcal{T}_i = \mathcal{O}_{\mathcal{X}}(x_i)$ and $0 \le a_i < e_i$. The round-down formula gives $\dim H^i(\mathcal{X}, \mathcal{L}) = \dim H^i(C, M)$ and $\deg_{\mathcal{X}} \mathcal{L} = \deg M + \sum_i \frac{a_i}{e_i}$. Nonzero fractional parts increase degree without changing cohomology. This proves both equalities and forces every minimizer to have trivial stacky characters. The remaining Betti rows may require finer invariants.

2.4 Toward a Strong Form of Stacky Green's Conjecture The strong form aims to recover the full Betti table from g, $\operatorname{Cliff}(C)$, and the signature; the PI has settled the simplest case.

Theorem 2.8 (One stacky point of order two)**.** *If* $\mathcal{X} = \sqrt[2]{(C, P)}$ *for* $P \in C$*, then for all* p, q*,*

$$\beta_{p,p+q}(R) = \beta_{p,p+q}\big(R(C, K_C)\big) + \beta_{p-1,p+q-2}\big(R(C, K_C)\big) + \delta_{q,2}\tbinom{g-1}{p} - \varepsilon_{p,q},$$

where $\varepsilon_{p,q} = 1$ *exactly for* $(p, q) \in \{(1, 1), (0, 2), (g-1, 2), (g-2, 3)\}$*.*

Thus Problem 2.4(2) has an affirmative answer here: $\beta(R)$ is independent of P. Indeed, $\mathcal{W}$ has weights $1^g, 2$, so $N = g + 1$, $c = g - 1$, and $\rho = 5$. Proposition 2.6 and Theorem 2.3 specify rows $0, 1, 3, 4$, leaving only row 2. In each degree j the Hilbert numerator, which depends only on the signature, determines the sole unknown $\beta_{j-2,j}$; for $\rho \ge 7$ this fails.

Conjecture 2.9 (Strong Stacky Green's Conjecture I)**.** *The Betti numbers* $\beta_{p,q}(\mathcal{X}; K_{\mathcal{X}})$ *depend only on* C *and the signature of* $\mathcal{X}$*, and not on the position of the stacky points.*

Proposition 2.6 and Theorem 2.3 confine positional dependence to the middle rows. I will test it through the connecting-map ranks below; rank drops would identify pointed invariants beyond the gonality and Clifford index fixed by Lemma 2.7.

The pole degeneration and our computed examples suggest the following bold conjecture for the full Betti table of a canonical stacky curve. For a graded S-module M, write $B_M(t, u) = \sum_{p,j} \beta^S_{p,j}(M) t^p u^j$. Let $J \subset S$ be the ideal generated by the variables of degree ≥ 2, and let P_C be the Betti polynomial of the coarse canonical ring over S/J. Set $B_J^\vee(t, u) = t^{c-1}u^{\sigma+1}B_J(t^{-1}, u^{-1})$.

Conjecture 2.10 (Strong Stacky Green's Conjecture II)**.** *We have*

$$B_{\mathcal{X}} = B_{S/J}\left(P_C + (1+tu)^{g-1}\sum_i \frac{u^2 + \cdots + u^{e_i}}{1 + tu^{e_i}}\right) - (1+t)(B_J + B_J^\vee).$$

The first term combines the coarse resolution with the higher-weight generators and the Koszul complexes on canonical forms vanishing at each P_i. The subtracted terms record consecutive cancellations paired by Gorenstein duality. To approach this formula, I will first compute the special-fiberesolution using (3), then control its cancellations under deformation.

Research Goal 2.11. *Prove that* $B_{\operatorname{gr}_F R}$ *equals the right-hand side of Conjecture 2.10 plus* $(1+t)B_J^\vee$*.*

Semicontinuity and Gorenstein duality would then bound. $\beta(R)$ by the conjectured table when $(1+t)B_J$ and $(1+t)B_J^\vee$ have disjoint support, as for $\kappa < g - 2$. In this range, proving equality reduces to excluding further cancellations under deformation. To control these cancellations geometrically, M. Banks, P. Wei, and I have developed a weighted analogue of Lazarsfeld's kernel-bundle method [109, 85], as suggested in [34, §7.3] . Set $\mathcal{L} = \mathcal{O}_{\mathcal{X}}(K_{\mathcal{X}})$ and define

$$0 \longrightarrow \mathcal{M}_{\mathcal{W}} \longrightarrow \mathcal{E}_{\mathcal{W}} \coloneqq \bigoplus_{d \ge 1} \mathcal{W}_d \otimes \mathcal{L}^{-d} \longrightarrow \mathcal{O}_{\mathcal{X}} \longrightarrow 0.$$

Exterior powers and cohomology give $\operatorname{Tor}_p^S(R, \mathbb{C})_j \cong \ker H^1\big(\wedge^{p+1}\mathcal{M}_{\mathcal{W}} \otimes \mathcal{L}^j \to \wedge^{p+1}\mathcal{E}_{\mathcal{W}} \otimes \mathcal{L}^j\big)$. I will begin with $\mathcal{X} = \sqrt[3]{(C, P)}$, the first signature beyond the order-two case, with weights $1^g, 2, 3$. Writing $\mathcal{T} = \mathcal{O}_{\mathcal{X}}(x)$, successive evaluation maps filter $\mathcal{M}_{\mathcal{W}}$ with factors $\pi^*M_{K_C} \otimes \mathcal{L}^{-1}$, $\mathcal{L}^{-2} \otimes \mathcal{T}^{-1}$, and $\mathcal{L}^{-3} \otimes \mathcal{T}^{-1}$. Exterior powers and pushforward reduce the calculation to twists of $\wedge^a M_{K_C}$ by

powers of K_C and multiples of P. The two extensions defining the filtration need not split; I will compute their connecting maps and combine them with the map to $\wedge^{p+1}\mathcal{E}_{\mathcal{W}}$ in the Tor formula. Here only rows $2, 3, 4$ remain undetermined, and duality exchanges rows 2 and 4, leaving $\beta_{p,p+2}$ and $\beta_{p,p+3}$ as the independent targets. For arbitrary signatures, I will express the differentials governing cancellation under the pole degeneration through the evaluation filtration and determine their ranks as the markings vary. Proving the prescribed ranks and ruling out additional cancellations would establish the full Betti formula for general markings; the loci where these ranks change would then show whether the formula extends to every marking or requires additional pointed invariants.

Schreyer's early work on Green's conjecture [126] remains, to my mind, a lovely demonstration of computation laying bare the geometry of syzygies. In that spirit, I propose to push further the computational tools that led to the conjectures and partial results above.

Research Goal 2.12. *Determine the generic Betti tables for signatures* $(g;3)$*,* $(g;4)$*, and* $(g;2,2)$ *with* $3 \leq g \leq 6$*, letting computation test and guide Conjectures 2.9 and 2.10.*

2.5 Beyond Stacky Canonical Curves The canonical case suggests a broader principle: coarse geometry controls syzygetic thresholds, while stacky points shift their homological position. I will test this for general section rings, addressing Brown and Erman's question on weighted N_p [34, Ques. 7.6]. Let $\mathcal{D} = \pi^*L \otimes \bigotimes_i \mathcal{T}_i^{a_i}$, where L is very ample on C, $0 < a_i < e_i$, $\gcd(a_i, e_i) = 1$, and $\mathcal{T}_i = \mathcal{O}_{\mathcal{X}}(x_i)$ as above, so $\deg \mathcal{T}_i = 1/e_i$, $\mathcal{T}_i^{e_i} \cong \pi^*\mathcal{O}_C(P_i)$, and $\lfloor \mathcal{D} \rfloor = L$. When $\deg L \geq 2g-1$, the coarse H^1 vanishings extend the pole filtration of Theorem 2.2 to $R(\mathcal{X}, \mathcal{D})$, replacing $b_i(d)$ by $\lfloor da_i/e_i \rfloor$ and T_{e_i} by the semigroup ring of $\{(n,\ell) : 0 \leq \ell \leq \lfloor na_i/e_i \rfloor\}$.

The first regime is $\deg L \geq 2g+1$, where classical vanishing theorems [86, 85] provide a starting point. Let $\mathcal{W}$ minimally generate $R(\mathcal{X}, \mathcal{D})$ and set $\kappa = \dim \mathcal{W} - h^0(C, L)$, the number of generators of weight greater than one. This depends on the stacky twist: $\kappa = \sum_i (e_i - 1)$ when every $a_i = e_i - 1$, but $\kappa = s$ when every $a_i = 1$. I predict that these generators shift the coarse threshold without otherwise changing it. As in §2.3, $\tau_{\mathrm{BE}}(\mathcal{X}, \mathcal{D})$ and $\tau(C, L)$ denote the first failures of weighted and ordinary N_p, respectively, with either threshold equal to ∞ if no failure occurs.

Conjecture 2.13. *If* $\deg L \geq 2g+1$*, then* $\tau_{\mathrm{BE}}(\mathcal{X}, \mathcal{D}) = \tau(C, L) + \kappa$.

I will begin with one stacky point of order e and $\mathcal{D} = \pi^*L \otimes \mathcal{T}$, so $\kappa = 1$. Write $R = R(\mathcal{X}, \mathcal{D})$. The evaluation filtration has factors $\pi^*M_L \otimes \mathcal{D}^{-1}$ and $\mathcal{D}^{-e} \otimes \mathcal{T}^{-1}$. In the first forbidden degree $j = p+e+1$, exterior powers and pushforward give $\wedge^{p+1}M_L \otimes L^e(P)$ and $\wedge^p M_L \otimes L$. Coarse regularity kills the first H^1, the target H^1 in the Tor formula, and the terms in higher forbidden degrees; the second is $K_{p-1,2}(C, L)$. The first target is therefore the comparison $\mathrm{Tor}_p^S(R, \mathbb{C})_{p+e+1} \cong K_{p-1,2}(C, L)$, proving the shift by one. For general κ, I will track the vanishing orders removed by successive evaluation maps and show that only the factor containing all higher-weight generators can contribute in forbidden degrees. This would recover the coarse Koszul group and explain the full shift. I will then test the method on log canonical rings, including modular-form rings with cusps, without using Gorenstein duality. These thresholds would complement recent positivity and syzygy results in nonstandard gradings [32, 31, 11, 115, 62].

At degree $2g$, secant geometry gives a sharper target. For $g = 2n+1$ and $\mathrm{Cliff}(C) = n$, Farkas–Kemeny show that a very ample, normally generated line bundle L of degree $2g$ satisfies N_{n-1} if and only if it is *n-very ample*: every subscheme of length $n+1$ imposes independent conditions [73, Thm. 1.4]; see also [72]. I predict that for general stacky points this criteria, shifted by κ still holds.

Conjecture 2.14 (Stacky Secant Conjecture)**.** *For C, L above, fixed $e_i \geq 2$, and general distinct $P_i \in C$, set $\mathcal{D} = \pi^*L \otimes \bigotimes_i \mathcal{T}_i^{e_i - 1}$, $\kappa = \sum_i (e_i - 1)$. $(\mathcal{X}, \mathcal{D})$ is weighted $N_{n-1+\kappa}$ iff L is n-very ample.*

The first test is genus five with one point of order two: weighted N_2 should distinguish general degree-ten line bundles from those with a length-three secant obstruction. Taking $L = K_C(D - Q)$, with D a general effective divisor of degree three and $Q \notin \mathrm{Supp}(D)$, gives an explicit family: L is nonspecial, but D imposes only two conditions. For arbitrary odd genus and signature, I will filter

the evaluation kernel one stacky generator at a time and seek a comparison of all groups forbidden by weighted $N_{n-1+\kappa}$ with shifted coarse Koszul groups. The key is to use the 2-regularity of $R(C, L)$ to eliminate the other H^1-terms, leaving Farkas–Kemeny's theorem to identify the surviving coarse obstruction with failure of n-very ampleness.

3. Multigraded Castelnuovo–Mumford Regularity: Bounds and Behavior

The PI linked cohomological vanishing and syzygies on products of projective spaces and proved finite-generation and asymptotic results for toric regularity (§1, [39, 40]). Yet regularity can have infinitely many corners, multigraded Betti numbers do not determine it, and geometric bounds remain scarce. I will explain its shape through support geometry, bound corners using syzygies, and bound curve regularity through intersection theory. Throughout, X, S, B, $B_1, \dots, B_s$ are as in §1, $n = \dim X$, and $\mathbb{K}$ is algebraically closed of characteristic zero.

3.1 The Shape and Minimal Degrees of Multigraded Regularity As the PI showed, the coordinate ring S/I_p of a point $p \in \mathcal{H}_2$ has infinitely many minimal regular degrees (Figure 1(c), [39]), although $\mathcal{O}_p$ is regular in every degree. These corners are measured relative to $\mathrm{Nef}(X)$, and I propose that their infinitude reflects the choice of cone: the ambient nef cone can be too small to account for the directions of positivity available to M. A cone adapted to M should recover persistence and finite generation, up to a uniform shift. The point example supports this expectation: its support-relative regularity has only one corner. This leads to three connected problems: determine the directions of regularity from geometry, bound its minimal degrees using syzygies, and control the number of corners. For a finitely generated graded S-module M, let $\mathcal{F} = \widetilde{M} \neq 0$ and $Z = \mathrm{supp}(\mathcal{F})$. With D_ρ the torus-invariant prime divisor of a ray ρ, set

$$K_Z = \bigcap_{p \in Z} \mathrm{cone}\{[D_\rho] : p \notin D_\rho\}, \qquad C_Z = \overline{\mathrm{cone}\{[C] : C \subset Z \text{ an integral curve}\}}^{\vee},$$

with curve classes in $N_1(X)_{\mathbb{R}}$. A rational class lies in K_Z when some positive multiple has an ambient linear system with no base point on Z, and in C_Z when its restriction to Z is nef; thus $\mathrm{Nef}(X)_{\mathbb{R}} \subset K_Z \subset C_Z$, where K_Z sees only which torus orbits meet Z and C_Z sees the Mori geometry of Z, toric or not. I conjecture that they are the recession cones of $\mathrm{reg}(M)$ and $\mathrm{reg}(\mathcal{F})$.

Conjecture 3.1. *If $H^0_B(M) = 0$, there are $\mathbf{a}_M, \mathbf{b}_M, \mathbf{a}_\mathcal{F}, \mathbf{b}_\mathcal{F} \in \mathrm{Pic}(X)$ such that*

$$\mathbf{a}_M + (K_Z \cap \mathrm{Pic}(X)) \subset \mathrm{reg}(M) \subset \mathbf{b}_M + K_Z \quad \textit{and} \quad \mathbf{a}_\mathcal{F} + (C_Z \cap \mathrm{Pic}(X)) \subset \mathrm{reg}(\mathcal{F}) \subset \mathbf{b}_\mathcal{F} + C_Z.$$

Module and sheaf regularity differ exactly by $H^1_B(M)$, the failure of sections to lift. In Figure 1(c), $\mathrm{reg}(S/I_p)$ fills $K_p \supsetneq \mathrm{Nef}(X)_{\mathbb{R}}$, while $C_p = \mathrm{Pic}(X)_{\mathbb{R}}$. For the diagonal $\Delta \subset \mathbb{P}^1 \times \mathbb{P}^1$, K_Δ is the nef cone but C_Δ is a half-plane: $\mathrm{reg}(S/(x_0y_1 - x_1y_0)) = \mathbb{N}^2$, while $\mathrm{reg}(\mathcal{O}_\Delta) = \{(a, b) : a + b \geq 0\}$, so one module corner shatters into infinitely many sheaf corners. I will approach the module bounds through toric modifications of linear systems whose base loci avoid Z. For sheaves, Fujita vanishing on Z [100, Thm. 1.5] and the global generation forced by regularity point to the two inclusions, and comparing first syzygies along support curves offers a route to effective translates [112].

Our faithfulness theorem (§1) gives finitely many corners but does not locate them, and when $K_Z \supsetneq \mathrm{Nef}(X)_{\mathbb{R}}$, Conjecture 3.1 forces infinitely many. I therefore measure corners relative to $\Gamma_Z \coloneqq K_Z \cap \mathrm{Pic}(X)$: let $\mathrm{reg}_Z(M)$ be Maclagan–Smith regularity [112] with respect to $B_1, \dots, B_s$ together with the minimal generators of Γ_Z, so that Γ_Z plays the role of $\mathrm{Nef}(X)$. Then $\mathrm{reg}_Z(M) \subseteq \mathrm{reg}(M)$ is Γ_Z-stable, with equality on products, and in Figure 1(c), $\mathrm{reg}_Z(S/I_p) = \Gamma_p$ has a single corner. I conjecture that, up to a uniform shift, regularity *is* support-relative regularity, with corners in a box determined by the Betti table.

Conjecture 3.2. *Fix X and a basis of $\mathrm{Pic}(X)$. There are $\mathbf{c}_X \in \mathrm{Nef}(X)$ and, for each multigraded Betti table β, an explicit finite box $Q_X(\beta)$ such that every finitely generated graded S-module M*

with $\beta^S(M) = \beta$, $\widetilde{M} \neq 0$, *and* $H^0_B(M) = 0$ *satisfies*

$$\mathbf{c}_X + \operatorname{reg}(M) \;\subseteq\; \operatorname{reg}_Z(M) \;=\; \big(Q_X(\beta) \cap \operatorname{reg}_Z(M)\big) + \Gamma_Z.$$

The inclusion gives persistence along Γ_Z, and the equality locates its corners; together they imply the module half of Conjecture 3.1. I will approach the inclusion by adapting the Maclagan–Smith nonzerodivisor argument to sufficiently divisible degrees in Γ_Z, tracking the nef shift through induction on $\dim Z$. For the box, Lauren Cranton Heller have proved the followng

Lemma 3.3. *Under the hypotheses of Conjecture 3.2, with* $X = \mathbb{P}^{n_1} \times \mathbb{P}^{n_2}$, *every corner of* $\operatorname{reg}(M)$ *lies in* $\prod_{j=1}^{2}[\ell_j(\beta), U_j(\beta)]$, *where* $\ell_j(\beta) = \min_{\beta_{0,\mathbf{a}} \neq 0} a_j$ *and* $U_j(\beta) = \max_{\beta_{i,\mathbf{a}} \neq 0} a_j$.

For $r \geq 3$, I will seek coordinatewise descent: if $\mathbf{d} \in \operatorname{reg}(M)$ and $d_j > U_j(\beta)$, then $\mathbf{d} - \mathbf{e}_j \in \operatorname{reg}(M)$; iterating would trap corners in a box. I will start with three factors, isolating one direction via partial Koszul complexes, then extend to faces of Γ_Z via diagonal resolutions [25, 49, 93, 36].

Corollary 3.4. *Assume Conjecture 3.2 and fix* β. *Among modules satisfying its hypotheses, only finitely many regions* $\operatorname{reg}_Z(M)$ *occur. Each is determined by* Γ_Z *and the finite set* $Q_X(\beta) \cap \operatorname{reg}_Z(M)$, *and* $\operatorname{reg}_Z(M) \subseteq \operatorname{reg}(M) \subseteq \operatorname{reg}_Z(M) - \mathbf{c}_X$.

This holds as Γ_Z depends only on which torus orbits meet Z, so only finitely many monoids occur; testing a corner still needs vanishing on all of Γ_Z. On products, $\Gamma_Z = \operatorname{Nef}(X)$ and the Maclagan–Smith criterion applies, so the $r = 2$ box theorem gives a complete algorithm for $\operatorname{reg}(M)$ on $\mathbb{P}^{n_1} \times \mathbb{P}^{n_2}$, towards [112, Prob. 4.10]; existing implementations, including ours [3], scan a fixed window and can miss corners outside it. In general, I will seek a finite criterion adapting the nonzerodivisor argument of [112, Cor. 4.8]. Our asymptotic theorem traps $\operatorname{reg}(I^p)$ between translates moving linearly in p, but corners may run superlinearly along faces.

Conjecture 3.5. *Let* $X = \mathbb{P}^{n_1} \times \cdots \times \mathbb{P}^{n_r}$, $I \subset S$ *be homogeneous, and* M *be a finitely generated graded* S*-module. There are constants* A, C *such that* $\|\mathbf{v}\|_\infty \leq Ap + C$ *for every corner* $\mathbf{v} \in \operatorname{reg}(I^p M)$.

The Betti degrees of $I^p M$ grow linearly in p [9, Thm. 1.1], so the proposed coordinatewise box on products would imply this when $H^0_B(M) = 0$; our $r = 2$ theorem already gives this case. For general M, I will separately bound the contribution of B-torsion using the Rees module. Since corners form an antichain, the window allows $O(p^{r-1})$ of them, and this is sharp: for the ideal $\mathfrak{m}$ of all Cox variables, $\operatorname{reg}(\mathfrak{m}^p) = \{\mathbf{d} \in \mathbb{N}^r : |\mathbf{d}| \geq p\}$ has $\binom{p+r-1}{r-1}$ corners. For ordinary powers I predict an arithmetic structure. Set $\mathcal{C}(I) \coloneqq \{(p, \mathbf{d}) \in \mathbb{N} \times \mathbb{Z}^r : p \geq 1,\ \mathbf{d} \text{ is a corner of } \operatorname{reg}(I^p)\}$, and call a set *semilinear* if it is a finite union of translates of finitely generated additive monoids.

Conjecture 3.6. *Let* $X = \mathbb{P}^{n_1} \times \cdots \times \mathbb{P}^{n_r}$ *with* $n_j \geq 1$, *and let* $0 \neq I \subsetneq S$ *be homogeneous. There is* p_0 *such that* $\mathcal{C}(I) \cap \{p \geq p_0\}$ *is semilinear.*

This would make the number of corners eventually quasipolynomial in p, of degree at most $r - 1$ [138]. I will first prove the monomial case. For monomial ideals, I will adapt constructibility under local cohomology [61, Thm. 8.1], then use Presburger quantification over fine degrees and nef shifts to describe regularity and its corners. In general, I will test Rees degenerations on $\mathbb{P}^1 \times \mathbb{P}^1$ and $\mathbb{P}^1 \times \mathbb{P}^2$, seeking conditions that prevent cohomological cancellation along the boundary. Even where corners change, this produces explicit bounds complementing the asymptotic sheaf regions of [41].

3.2 A Bound for Multigraded Regularity in the Style of Gruson, Lazarsfeld, and Peskine A foundational result of Gruson, Lazarsfeld, and Peskine [89] states that if $C \subset \mathbb{P}^r_{\mathbb{K}}$ is an integral nondegenerate curve defined over an algebraically closed field $\mathbb{K}$, then $\operatorname{reg}(I_C) \leq \deg(C) - r + 2$, where $I_C \subset \mathbb{K}[x_0, \ldots, x_r]$ is the (saturated) homogeneous defining ideal of C. Since $\deg(C) = C \cdot H$ for the hyperplane class H, the theorem is a strong bridge between the intersection theory of C and the degrees of its equations and syzygies. It sharpens classical work of Castelnuovo [45] on space curves and settles the curve case of the Eisenbud–Goto conjecture $\operatorname{reg}(I_C) \leq \deg(C) - \operatorname{codim}(C) + 1$

[68]. It has inspired decades of work on optimal bounds [124, 108, 105, 106, 82] and on bounding regularity by the degrees of defining equations [19, 66, 60, 111], and recent counterexamples to the general Eisenbud–Goto conjecture [116, 46, 117] make such geometric bounds all the more valuable.

Research Problem 3.7. *Prove a multigraded analogue of Gruson, Lazarsfeld, and Peskine's (GLP) theorem for curves in smooth projective toric varieties.*

I seek an explicit region in $\operatorname{reg}(\mathcal{I}_C)$ shaped by X's divisor relations [112]. Cobb adapted the kernel bundle techniques of GLP to products of projective spaces [89, 52], and Brown–Erman ask for a weighted analogue [34, Ques. 7.5]. Classically, $0 \to \mathcal{I}_C \to \mathcal{O}_{\mathbb{P}^r} \to \mathcal{O}_C \to 0$ reduces m-regularity of $\mathcal{I}_C$ to three tests. (i) *Ambient* ($q \geq 3$): $H^q(\mathcal{I}_C(m-q)) \cong H^q(\mathcal{O}_{\mathbb{P}^r}(m-q)) = 0$, since $H^{q-1}(\mathcal{O}_C(m-q)) = 0$ and $\mathbb{P}^r$ has no intermediate cohomology. (ii) *Speciality* ($q = 2$): $H^2(\mathcal{I}_C(m-2)) \cong H^1(\mathcal{O}_C(m-2))$ vanishes for nonspecial $\mathcal{O}_C(m-2)$, and $(m-2)\deg(C) > 2p_a(C)-2$ suffices, holding at $m = \deg(C) - r + 2$ by Castelnuovo's genus bound. (iii) *Interpolation* ($q = 1$): $H^1(\mathcal{I}_C(m-1))$ is the cokernel of $H^0(\mathcal{O}_{\mathbb{P}^r}(m-1)) \to H^0(\mathcal{O}_C(m-1))$, vanishing exactly when sections on C extend to $\mathbb{P}^r$. Kernel bundles prove this surjectivity. Nondegeneracy gives $\operatorname{length}(C \cap \ell) \leq \deg(C) - r + 2$ for every secant line ℓ; interpolation on an s-secant line forces $\operatorname{reg}(\mathcal{I}_C) \geq s$, so maximal secants govern the bound.

Now let X be as in this section with $n \geq 2$. Say that an integral curve $C \subset X$ is *nondegenerate* if each restriction $H^0(X, B_j) \to H^0(C, B_j|_C)$ is injective. Set $r_j \coloneqq \dim_{\mathbb{K}} H^0(X, B_j) - 1$, $d_j \coloneqq \langle B_j \cdot C \rangle$, and the *defect* $\delta_j \coloneqq d_j - r_j$. Maclagan–Smith regularity of $\mathcal{I}_C$ at $\mathbf{d}$ tests the twists $\mathbf{d} - B_i$, then $\mathbf{d} - B_i - B_j$, and so on, so the same three questions arise. I propose answering each by a region depending only on X and the numerical class of C: (i) *Ambient.* Membership in $\operatorname{reg}(\mathcal{O}_X)$ gives $H^q(X, \mathcal{O}_X(\mathbf{d} - \sum_j v_j B_j)) = 0$ for $q > 0$ and $|\mathbf{v}| = q$. Since C is a curve, for $q \geq 3$ these are exactly the ideal-sheaf tests. (ii) *Speciality.* Choose an explicit numerical bound $\widehat{G}_X([C]) \geq p_a(C)$. For example, $A = \sum_j B_j$ is very ample, and multiplication of the restricted section spaces shows that C spans at least a $\mathbb{P}^R$, where $R = \sum_j r_j$. Thus one may take Castelnuovo's bound $\pi(D, R)$ [6, Ch. III], where $D = \sum_j d_j$: writing $D - 1 = m(R-1) + \epsilon$ with $0 \leq \epsilon < R - 1$, this is $\binom{m}{2}(R-1) + m\epsilon$. On a surface, adjunction gives the exact genus and can be used instead. Every second twist is nonspecial on $\mathcal{H}_X([C]) \coloneqq \{\mathbf{d} \in \operatorname{Pic}(X) : \mathbf{d} \cdot C \geq 2\widehat{G}_X([C]) - 1 + 2\max_j d_j\}$, because $\deg \mathcal{O}_C(\mathbf{d} - B_i - B_j) > 2p_a(C) - 2$. (iii) *Interpolation.* Guided by the secant-line explanation above, set $M_X \coloneqq \overline{\operatorname{NE}}(X) \cap N_1(X)_{\mathbb{Z}}$ and $e_j(\gamma) \coloneqq B_j \cdot \gamma$. Call $0 \neq \gamma \in M_X$ *eligible* if $e_j(\gamma) < r_j$ for some j, and put $q_C(\gamma) \coloneqq \min_{j:\, e_j(\gamma) < r_j}(\delta_j + e_j(\gamma))$ and $w(\gamma) \coloneqq \max_j e_j(\gamma)$. If an integral rational curve $C' \neq C$ has eligible class γ, its image under B_j spans at most $\mathbb{P}^{e_j(\gamma)}$. The resulting sections vanishing on C' and nondegeneracy of C give $\operatorname{length}(C \cap C') \leq q_C(\gamma) + 1$. When C' is smooth, any values on a subscheme of length $q_C(\gamma) + 1$ are interpolated by line bundles of degree $\geq q_C(\gamma)$. Requiring this degree at every first twist motivates:

$$\mathcal{S}_X([C]) \coloneqq \{\mathbf{d} \in \operatorname{Pic}(X) \mid \mathbf{d} \cdot \gamma \geq q_C(\gamma) + w(\gamma) \text{ for every eligible } \gamma \in M_X\}.$$

Conjecture 3.8 (Multigraded GLP)**.** *If $C \subset X$ is an integral nondegenerate curve, then*

$$\mathcal{R}_X([C]) \coloneqq \mathcal{S}_X([C]) \cap \mathcal{H}_X([C]) \cap \operatorname{reg}(\mathcal{O}_X) \;\subseteq\; \operatorname{reg}(\mathcal{I}_C).$$

On $\mathbb{P}^r$, the line class gives $m \geq \deg(C) - r + 2$, higher eligible multiples give weaker inequalities, and Castelnuovo's genus bound supplies nonspeciality at the same threshold, so the conjecture recovers the classical theorem. I have proved it for curves in $\mathbb{P}^{n_1} \times \mathbb{P}^{n_2}$ with a minimal-degree projection, and computations for curves in $\operatorname{Bl}_p \mathbb{P}^3$ have found no counterexamples. The bound is sharper than Cobb's even for a simple curve: the image $C \subset X = \mathbb{P}^2 \times \mathbb{P}^2$ of $[u : v] \mapsto \big([u^3 : u^2v : v^3], [u^2 : uv : v^2]\big)$, which pairs the normalization of a cuspidal cubic with the conic embedding and is therefore smooth, with multidegree $(3, 2)$ and defects $(1, 0)$. The inequalities defining $\mathcal{S}_X([C])$ reduce to $a \geq 1$ and $b \geq 2$; the genus estimate $\widehat{G}_X([C]) = \pi(5, 4) = 1$ adds only the redundant $3a + 2b \geq 7$. In fact,

$\mathcal{R}_X([C]) = \operatorname{reg}(\mathcal{I}_C) = (1,2) + \mathbb{N}^2$, so the bound is sharp (Lozovanu also obtains this corner over $\mathbb{C}$ [110, Thm. A]), whereas Cobb's bound gives $(3,3) + \mathbb{N}^2$ [52, Prop. B].

What remains is precise: the ambient and nonspeciality conditions give the other regularity tests, so it suffices to prove that $H^0(X, \mathcal{O}_X(\mathbf{d} - B_i)) \to H^0(C, \mathcal{O}_C(\mathbf{d} - B_i))$ is surjective for all $\mathbf{d} \in \mathcal{R}_X([C])$ and i. For general X, I will combine the classical auxiliary line bundle on the normalization $f\colon \widetilde{C} \to X$ with the Hanlon–Hicks–Lazarev diagonal [89, 52, 93], whose terms are $\mathcal{D}_p = \bigoplus_\lambda \mathcal{O}_X(-A_{p\lambda}) \boxtimes \mathcal{O}_X(-E_{p\lambda})$. For a line bundle L on $\widetilde{C}$, pulling back, tensoring, and pushing forward give a spectral sequence with $E_1^{-p,q} = \bigoplus_\lambda H^q\big(\widetilde{C}, L \otimes f^*\mathcal{O}_X(-A_{p\lambda})\big) \otimes \mathcal{O}_X(-E_{p\lambda})$ for $q = 0, 1$, converging to f_*L in total degree zero. I will first use Brown–Sayrafi's diagonal on $\operatorname{Bl}_p \mathbb{P}^3$ [36]. For general L, the inequalities $\deg L - A_{p\lambda} \cdot C \geq g(\widetilde{C}) - 1$ kill the relevant H^1. I will compute the resulting presentation and test whether its determinantal complex yields first-twist surjectivity throughout $\mathcal{R}_X([C])$, beginning with smooth curves. Should extra positivity be needed, I will obtain an explicit weaker region and identify the missing sharpening. For singular curves, I will track the normalization quotient; curves in $\mathbb{P}^1 \times \mathbb{P}^2$ are a first graduate-student project.

3.3 Multigraded Regularity of Torus-Invariant Subvarieties On a smooth projective toric variety $X = X_\Sigma$, the torus-invariant irreducible subvarieties are in dimension-reversing bijection with the cones of Σ. If $S = \mathbb{K}[x_\rho \mid \rho \in \Sigma(1)]$ is the Cox ring, the subvariety Z_σ corresponding to $\sigma \in \Sigma$ has Cox ideal $I_{Z_\sigma} = \langle x_\rho \mid \rho \in \sigma(1) \rangle$, which is resolved by a multigraded Koszul complex. Even so, the multigraded regularity of these subvarieties can be subtle: the four torus-fixed points of the Hirzebruch surface $\mathcal{H}_2$ exhibit two distinct regularities, one with infinitely many minimal corners [39]. These highly structured subvarieties are thus a natural testing ground for how regularity reflects positivity beyond syzygies, yet no concrete description of their regularity is known.

Research Problem 3.9. *Let $Z_\sigma \subset X$ be an irreducible torus-invariant subvariety corresponding to the cone σ. Give an explicit combinatorial characterization of* $\operatorname{reg}(I_{Z_\sigma})$ *and* $\operatorname{reg}(S/I_{Z_\sigma})$.

We ask for both because, unlike in the standard graded case, $\operatorname{reg}(S)$ need not have a single corner, so the two can behave quite differently [39]. Our approach combines Hilbert-function combinatorics with explicit higher cohomology of line bundles on torus-invariant subvarieties. With Jacob Lehman Duke, I have settled torus-fixed points. For a maximal cone σ, the variables surviving in S/I_{Z_σ} are indexed by rays outside σ, and the degrees of surviving monomials form the monoid $Q_\sigma = \{\sum_{\rho \notin \sigma(1)} a_\rho [D_\rho] \mid a_\rho \in \mathbb{N}\} \subset \operatorname{Pic}(X)$. By smoothness, the classes $[D_\rho]$ for $\rho \notin \sigma(1)$ form a lattice basis of $\operatorname{Pic}(X)$, so the Hilbert function of S/I_{Z_σ} is 1 on Q_σ and 0 elsewhere.

Proposition 3.10. *If $Z_\sigma \subset X$ is a torus-fixed point corresponding to a maximal cone $\sigma \in \Sigma$, then* $\operatorname{reg}(S/I_{Z_\sigma}) = Q_\sigma$ *and* $\operatorname{reg}(I_{Z_\sigma}) = \operatorname{reg}(S) \cap \bigcap_{j=1}^{s} (B_j + Q_\sigma)$.

Since S/I_{Z_σ} has no higher sheaf cohomology, its regularity is the set of degrees where Hilbert function and polynomial agree [112, Prop. 6.4], namely Q_σ. A cohomology argument reduces $\operatorname{reg}(I_{Z_\sigma})$ to surjectivity of evaluation at each first twist, i.e. $\mathbf{d} - B_j \in Q_\sigma$ for all j, giving the intersection. $\operatorname{reg}(S/I_{Z_\sigma})$ depends only on σ and the rays of Σ, so it is shared by all toric models with these data. Next are invariant curves $Z_\sigma \cong \mathbb{P}^1$, whose sheaf regularity is the half-space $\mathbf{d} \cdot Z_\sigma \geq \max_j (B_j \cdot Z_\sigma) - 1$, so I will determine which sections surviving Cox monomials represent. In higher dimension, I will combine these character counts with the toric Čech complex of the star of σ.

Such descriptions would also make the degeneration bounds of Maclagan and Smith explicit: they bound the regularity of a monomial quotient by intersecting shifted regularity regions of the factors in a Stanley filtration, which for saturated monomial ideals are coordinate rings of orbit closures [113, Thm. 4.1]. Answering Problem 3.9 combinatorially would compute these from the fan, and semicontinuity carries the bounds from initial ideals to the quotient [112, Prop. 6.13].

4. Uniform Homological Algebra: Regularity on the Cox Category

I propose a deliberately speculative test: *multigraded regularity may be asking the right questions, but in the wrong category.* Work of the PI and others shows that the classical link between cohomological vanishing and syzygies is fragile [112, 40, 39]: even the Cox ring of the Hirzebruch surface $\mathcal{H}_2$, free over itself, is not $\mathbf{0}$-regular [112, Ex. 4.3]. Uniformity across toric birational models is a growing theme. It begins with the Hanlon–Hicks–Lazarev resolution of the diagonal, which depends only on the rays of the fan [93], and extends to short resolutions, cellular constructions, and derived categories [33, 76, 77, 78, 57, 23, 17, 92, 70]. King's conjecture, that every smooth projective toric variety has a full strong exceptional collection of line bundles, is false [95, 118]; Ballard et al. recover it by passing to the *Cox category*, which sees all birational models at once [10]. I will use this uniform diagonal to define regularity where the structure sheaf is $\mathbf{0}$-regular on every smooth model.

4.1 The Cox Category and Diagonal Heights Let X be a projective normal toric variety over an algebraically closed field $\mathbb{K}$ of characteristic zero, with Cox ring S. The maximal GKZ chambers give models X_i and smooth toric stacks $\mathcal{X}_i$, with $\mathcal{X}_i = X_i$ when X_i is smooth. Following [10]. choose a common smooth toric stack with proper birational maps $\pi_i \colon \widetilde{\mathcal{X}} \to \mathcal{X}_i$ and set $D_{\mathrm{Cox}}(X) \coloneqq \langle \pi_i^* D^b(\mathcal{X}_i) \rangle \subseteq D^b(\widetilde{\mathcal{X}})$. Here $D^b(-) = D^b(\mathrm{Coh}(-))$, and all functors are derived. The category is independent of this choice and contains each $D^b(\mathcal{X}_i)$ fully faithfully. The divisor-class map $Q \colon \mathbb{Z}^{\Sigma(1)} \to \mathrm{Cl}(X)$ determines the half-open zonotope $Z_X = Q_{\mathbb{R}}((-1,0]^{\Sigma(1)})$ and the finite *Bondal–Thomsen collection* Θ of classes mapping into Z_X [133, 21]. Pulling back $\mathcal{O}(\alpha)$ from a chamber containing $-\alpha$ gives objects $T_\alpha \coloneqq \mathcal{O}_{\mathrm{Cox}}(\alpha)$ forming a full strong exceptional collection [10].

The uniform HHL complex lifts to a diagonal kernel inducing the identity on $D_{\mathrm{Cox}}(X)$ [10, § 6]. Take its minimal representative in the additive category generated by the fixed external products $T_\alpha \boxtimes T_\beta$, with terms $D_p = \bigoplus_{\alpha,\beta\in\Theta} (T_\alpha \boxtimes T_\beta)^{m_{p,\alpha,\beta}}$ for $p \geq 0$. Use the first factor as cohomological input, with $p = 0$ adjacent to the augmentation. For the occurring input labels $\alpha \in \Theta_H$, set $h(\alpha) \coloneqq \min\{p : m_{p,\alpha,\beta} \neq 0 \text{ for some } \beta\}$ and $\nu(\alpha) \coloneqq \max\{1, h(\alpha)\}$. The *height* records first occurrence in the minimal diagonal, not necessarily face codimension in the zonotope.

Definition 4.1. *Call* $\mathcal{E} \in D_{\mathrm{Cox}}(X)$ $\mathbf{0}$-(Cox) regular *if* $\mathrm{Ext}^q(T_\alpha^\vee, \mathcal{E}) = 0$ *for* $\alpha \in \Theta_H$, $q \geq \nu(\alpha)$.

Degree-q cohomology in diagonal degree p contributes in position $p - q$. The cutoff leaves only positive positions at positive height and allows sections at height zero. On $\mathbb{P}^n$, $h(-j) = j$ for $0 \leq j \leq n$, recovering classical 0-regularity. Regularity relative to an exceptional collection goes back to Costa–Miró-Roig [54, 55]; the new feature is the height cutoff, uniform across models.

Research Problem 4.2. *Find a combinatorial formula for* $h(\alpha)$ *from the zonotope and GKZ fan.*

Starting from the Betti-number formulas of Favero–Sapronov and Han [78, 92], I will express Cox heights directly through zonotope geometry and GKZ wall identifications. In Figure 3, two labels of different heights become the same line bundle on the weighted projective stack.

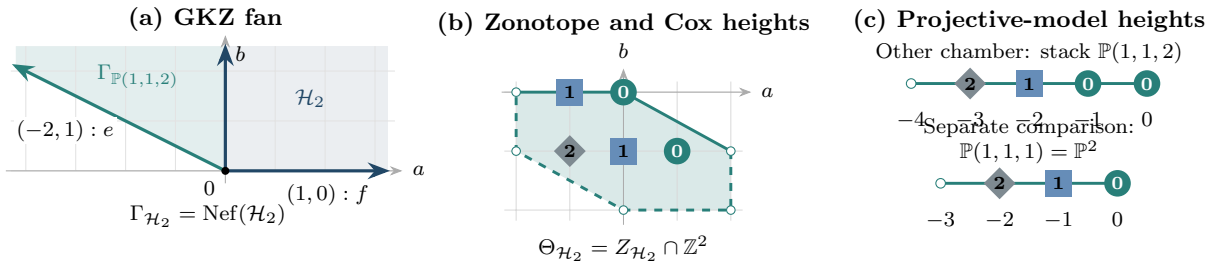


FIGURE 3. Diagonal heights for $\mathcal{H}_2$. (a) GKZ chambers; contracting e gives $\mathbb{P}(1,1,2)$. (b) Heights from the diagonal of [57, §4]; dashed edges and open vertices are excluded. (c) Heights on $\mathbb{P}(1,1,2)$ and on $\mathbb{P}^2$.

4.2 The Structure Sheaf and Nef Geometry The first test of Definition 4.1 is whether it repairs the failure on $\mathcal{H}_2$ that motivated it. To compare with classical regularity on a single model, fix a smooth projective model X_i and, for a coherent sheaf $\mathcal{F}$ on X_i, let $\mathcal{R}_i(\mathcal{F}) \subseteq \mathrm{Pic}(X_i)$ be the set of $\mathbf{d}$ for which $\pi_i^*(\mathcal{F}(\mathbf{d}))$ is $\mathbf{0}$-regular. The Θ-Transform Lemma [10, Lemma 1.6] and the projection formula translate the definition into ordinary sheaf cohomology on X_i: $\mathrm{Ext}^q(T_\alpha^\vee, \pi_i^*(\mathcal{F}(\mathbf{d}))) \cong H^q(X_i, \mathcal{F}(\mathbf{d}+\alpha))$. Thus $\mathbf{d} \in \mathcal{R}_i(\mathcal{F})$ exactly when $H^q(X_i, \mathcal{F}(\mathbf{d}+\alpha)) = 0$ for all $\alpha \in \Theta_H$ and $q \geq \nu(\alpha)$, a finite list of vanishing conditions indexed by the zonotope.

Lemma 4.3. *The object $\mathcal{O}_{\mathrm{Cox}}$ is $\mathbf{0}$-regular, and $\mathrm{Nef}(X_i) \subseteq \mathcal{R}_i(\mathcal{O}_{X_i})$ on every smooth model X_i.*

Indeed, Bondal–Thomsen vanishing gives $H^q(\mathcal{X}_i, \mathcal{O}_{\mathcal{X}_i}(\alpha)) = 0$ and $H^q(X_i, \mathcal{O}_{X_i}(\mathbf{d}+\alpha)) = 0$ for $q > 0$, $\alpha \in \Theta$, and nef $\mathbf{d}$ [10, Thm. 2.11]. Since $\nu(\alpha) \geq 1$ and $\mathcal{O}_{\mathrm{Cox}}$ is the common pullback of the structure sheaves, both claims follow. So, unlike Maclagan–Smith regularity, Cox regularity makes the structure sheaf $\mathbf{0}$-regular on every model, as on $\mathbb{P}^n$. The lemma uses only $\nu \geq 1$; the heights enter in the converse: does Cox regularity keep the positivity consequences of classical regularity?

Question 4.4. *Let $\mathcal{F}$ be a coherent sheaf on a smooth projective model X_i. Is $\mathcal{R}_i(\mathcal{F}) + \mathrm{Nef}(X_i) \subseteq \mathcal{R}_i(\mathcal{F})$? Is $\mathcal{F}(\mathbf{d})$ globally generated for every $\mathbf{d} \in \mathcal{R}_i(\mathcal{F})$?*

Since globally generated line bundles are nef, global generation would give $\mathcal{R}_i(\mathcal{O}_{X_i}) = \mathrm{Nef}(X_i)$. To prove persistence, I will resolve $\mathcal{O}_{X_i}(\alpha + B_j)$, for Hilbert-basis generators B_j of $\mathrm{Nef}(X_i)$, by sums of test line bundles $\mathcal{O}_{X_i}(\beta)$ with $\nu(\beta) \leq \nu(\alpha) + p$ in homological degree p; the cohomology spectral sequence then gives vanishing, and iteration handles all nef translates. I will construct these finite certificates first on Hirzebruch surfaces, then on scrolls. For global generation, the height cutoff leaves only diagonal-degree-zero sections in position zero; I will identify their output bundles and show that their evaluation maps have globally generated images.

The decisive test is multiplication of sections. On $\mathbb{P}^n$, Mumford's theorem makes multiplication by sections surjective above the regularity of a locally free sheaf. Maclagan–Smith regularity keeps this [112], but at the cost seen on $\mathcal{H}_2$: nef twists of $\mathcal{O}$ need not be regular. Cox regularity reverses the trade-off, making it a test of *Oda's conjecture*: on a smooth projective toric variety, $H^0(\mathcal{O}(\mathbf{d})) \otimes H^0(\mathcal{O}(A)) \to H^0(\mathcal{O}(\mathbf{d}+A))$ is surjective for $\mathbf{d}$ nef and A ample [120]. It is known for surfaces [71] and for threefolds up to finitely many ample classes [139], but open in general; with $\mathbf{d} = kA$ it would make every smooth lattice polytope normal, itself open in dimension three [14].

Conjecture 4.5 (Cox–Mumford Multiplication)**.** *Let $\mathcal{F}$ be locally free on a smooth projective model X_i, and let $\mathbf{d} \in \mathcal{R}_i(\mathcal{F})$. For every ample Cartier divisor A on X_i, the multiplication map $H^0(X_i, \mathcal{F}(\mathbf{d})) \otimes H^0(X_i, \mathcal{O}_{X_i}(A)) \to H^0(X_i, \mathcal{F}(\mathbf{d}+A))$ is surjective.*

By Lemma 4.3, the case $\mathcal{F} = \mathcal{O}_{X_i}$ implies Oda's conjecture. So the provocation is testable: if $D_{\mathrm{Cox}}(X)$ is the right category, Mumford's argument there should prove Oda's conjecture; a counterexample would show where the heights mismeasure positivity. As modest evidence towards this granted bold conjecture, I have proved it for Hirzebruch surfaces via a mapping cone argument

Lemma 4.6. *Conjecture 4.5 holds for every Hirzebruch surafce.*

I will extend the mapping-cone construction to split projective bundles, combining base and fiber evaluation sequences [114]. For general smooth toric models, the target is a finite resolution of the evaluation kernel $\mathcal{M}_A$ by test bundles $\mathcal{O}_{X_i}(\beta)$ with $\nu(\beta) \leq p+1$ in homological degree p, uniformly in A. Cox regularity would then force $H^1(X_i, \mathcal{F}(\mathbf{d}) \otimes \mathcal{M}_A) = 0$, proving Conjecture 4.5. The Cox diagonal applied to $\mathcal{M}_A$ gives such a resolution except for degree-zero terms $T_\alpha \boxtimes T_\beta$ with $\alpha \neq 0$ (e.g. the label $(1,-1)$ in Figure 3), which reintroduce multiplication maps and carry Oda's difficulty. When 0 is the only height-zero label, as on products of projective spaces, the argument should go through; I will prove this case first. Intermediate goals are Problem 4.2, Question 4.4 for Hirzebruch surfaces and scrolls, and a computer search for counterexamples on smooth toric

5. Extending Computational Algebra Software

Computation drives recent progress in multigraded commutative algebra and toric geometry. Building on the PI's track record, each project yields a community tool supporting this proposal.

5.1 The `toricStacks` Package for *Macaulay2* A lack of software is a major bottleneck for homological invariants of stacks. A general framework is out of reach, but toric stacks correspond to *stacky fans*, whose well-developed combinatorics makes computation feasible [80, 81, 24]; yet no implementation exists. Given the success of *NormalToricVarieties* [128], a *toricStacks* package would benefit the community and directly support the experiments in §2.

Research Problem 5.1. *Develop a Macaulay2* [84] *package for computations with toric stacks.*

The package will (i) construct toric stacks from stacky fans and from families such as weighted projective stacks and root stacks, (ii) compute the Cox ring with its $\mathrm{Cl}(\mathcal{X})$-grading and irrelevant ideal, and (iii) support coherent sheaves, including sheaf cohomology via local cohomology [69]. This is joint work in progress with Maya Banks and John Cobb; (i) is complete and (ii) is underway.

5.2 A Database of Human-Generated Examples in Algebra Good examples drive conjectures, theorems, and algorithms. Thousands of expert-chosen rings, ideals, and modules lie scattered, largely untapped, across *Macaulay2*'s open-source documentation and tests.

Research Problem 5.2. *Build a Macaulay2 package collecting the algebraic objects in its source and documentation into a queryable database with provenance and invariants.*

It will parse and rerun installed packages' `EXAMPLE`/`TEST` blocks in a sandbox, recording directly callable objects with source and authorship. Much of the parser exists; execution and curation remain. This conjecture testbed and benchmark suite meets growing demand for mathematician-built AI/ML on mathematician-owned, fully attributed data [53], complementing curated repositories [132, 79].

6. Broader Impacts

6.1 Organizing. Building on Section 1.5, since 2022 I have organized *Gems of Combinatorics II* (30 participants), *AGNES* (150), and *BATMOBILE 2024 & 2025* (50/30), and I co-organize Dartmouth's *Algebra & Number Theory* seminar. *Future Plans:* With co-organizers, I have proposed a *Thematic Semester in Combinatorial Algebraic Geometry* to the Fields Institute for 2030. I will host the next *Gems of Commutative Algebra* at Dartmouth in Fall 2028 and write a *Notices of the AMS* article on organizing early-career conferences. John Voight and I are in the planning stages of a Macaulay2+Magma workshop on the future of mathematical software.

6.2 Open Educational Resources. In Fall 2026 I am lecturing for the Preliminary Arizona Winter School, a nine-week virtual course with 300 students (200 undergraduate and master's students); all materials (videos and 100 pages of IBL notes and problems) are freely available. Building on the work of Karen Smith, I designed an inquiry-based commutative algebra course of 16 worksheets (200+ pages). Both are on my website. *Future Plans:* I will develop these materials into an open-access introduction to algebraic geometry with Macaulay2 exercises.

6.3 Open Computational Resources. I created *Syzygy Data*, `syzygydata.com`, a public database of large syzygy computations, and co-authored four Macaulay2 packages. *Future Plans:* I will lead a group developing `toricStacks` at the 2027 *Macaulay2* conference at Auburn.

6.4 Campus Community. I am the faculty advisor to Dartmouth's *AWM Student Chapter* and work with the campus *oSTEM* chapter to broaden participation in STEM. In Fall 2025 I founded *Unwritten Rules*, a graduate seminar that makes the hidden curriculum of an academic career explicit. *Future Plans:* I will continue *Unwritten Rules* and extend it to Dartmouth postdocs.

6.5 National & International Service. I have worked with the *AWM* Executive Committee to expand support for a broader range of mathematicians. From Winter 2023 until its dissolution in Spring 2025, I served on SLMath's *Committee on Women in Mathematics*. I now serve on the *AMS Committee on Engagement, Participation, & Advancement* and the *AMS Simons Travel Grant Committee*. In January 2026 I began a three-year term on the Board of Trustees of *Spectra*.

6.6 Mentoring. I advise three Ph.D. students on problems central to this proposal: Will D. (algebraic geometry + AI/ML), Jacob L-D. (regularity of torus-invariant subvarieties), and Alex M. (positivity of the BT collection). At MREG 2023 I began mentoring two graduate students on a project that led to [37]. In Summer 2026 I mentored three undergraduates through *Count Me In* (organized by M. Hering), and since Fall 2024 I have mentored two undergraduates. *Future Plans:* I will organize a summer undergraduate research program through Dartmouth's SHUR program.

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